

3-Tier Data Provenance & Quantitative Confidence System

Every station and planning zone carries an immutable data quality label and confidence score

TIER 1: LIVE

Real-Time Telemetry & OCPP

- Direct OCPP 1.6 WebSocket hardware
- Active port occupancy & power flow
- Verified ground-truth port state

Confidence Score: 0.95 – 1.00

TIER 2: ESTIMATED

Spatial API Fusion & Queues

- OpenChargeMap, Google Places, OSM
- M/M/c Erlang C queueing estimates
- Gradient Boosted demand forecasts

Confidence Score: 0.70 – 0.94

TIER 3: FALLBACK

Deterministic Heuristics

- Seeded deterministic random signals
- Zero unhandled rate-limit failures
- Explicit fallback_reason field

Confidence Score: 0.30 – 0.69